

1 The avg battery life of smartphone is 12 hrs with population standard deviation 2hr. A random sample of 64 phones is selected. estimate the standard error of the sample mean.

standard error Example mean:

Given

$$\sigma = 2, n = 64$$

formula

$$SE = \frac{\sigma}{\sqrt{n}}$$

$$SE = \frac{2}{8} = 0.25$$

$$\text{Ans} = 0.25 \text{ hrs}$$

2 Binomial poisson distribution

binomial distribution fixed
no. of trials

Poisson distribution
cont. exists in a
times space interval

Eg: Emails elicited, Co not
elicited

Eg: number of website
visits per hour.

data science applications:-

- binomial: email spam detection
- poisson: no. of customer request per

Importance of co-variance matrix

- 1) Contains variance of variables
- 2) shows Covariance b/w variables
- 3) describes relation ship among variables
- 4) used in multivariate normal distribution
- 5) helps in machine learning and PCA

Bernoulli random variable

$$p = 0.93, q = 0.07$$

4)

Variance

$$\text{Var}(X) = pq$$

$$= 0.93 (0.07)$$

$$= 0.0651$$

standard deviation

$$\sigma = \sqrt{0.0651}$$

$$= 0.255$$

$$\mu = 0.93, \text{Var}(X) = 0.051, \sigma = 0.2$$

5)

geometric distribution

$$p = 0.15$$

$$q = 0.75$$

probability of first click on 5th impression

$$P(X=5) = q^4 p$$

$$= (0.75)^4 (0.15)$$

$$= 0.0783$$

Expected no. of impressions

$$P_2(X) = \frac{1}{p}$$

$$= \frac{1}{0.15}$$

$$= 6.67$$

$$P(X=5) = 0.783, E(X) = 6.67$$

Poisson

distribution

$$\lambda = 8$$

exactly 6 Hokebs

$$P(X=6) = \frac{e^{-8} (8)^6}{6!}$$

$$\approx 0.1221$$

more than 8 Hokebs

$$P(X > 8) = 1 - P(X \leq 8)$$

$$= 1 - 0.5925$$

$$= 0.4075$$

Central limit theorem

$$\mu = 4.50$$

$$\sigma = 60, \quad n = 36$$

standard

$$SE = \frac{60}{\sqrt{36}}$$

$$SE = \frac{60}{6}$$

$$= 10$$

Probability sample mean exceeds

$$Z = \frac{470 - 450}{10}$$

$$(Z > 2) = 1 - 0.9772$$

$$= 0.0228$$

$$SE = 10, \quad P(\bar{x} > 470) = 0.0228.$$

8)

Sample distribution.

$$\mu = 4.5 \quad \sigma = 1.2 \quad n = 64$$

Standard Error

$$SE = \frac{1.2}{8}$$

$$= 0.15$$

Probability

$$z = \frac{4.8 - 4.5}{0.15}$$

$$= 2$$

$$P(z > 2) = 0.9772$$

$$SE = 0.15 \quad P(\bar{x} < 4.8) = 0.9772$$

9) Central limit theorem

$$\mu = 350$$

$$\sigma = 120$$

$$n = 4$$

$$SE = \frac{120}{8}$$

$$= 15$$

Probability

$$z =$$

$$\frac{375 - 350}{15}$$

$$= 1.67$$

$$P(z > 1.67) = 1 - 0.9525$$

$$= 0.0475$$

Applicability

$$\text{since } n = 64 > 30$$

CLT applicable

$$SE = 15, \quad P(\bar{x} > 375) = 0.0475$$

10) Covariance matrix

$$\Sigma = \begin{bmatrix} 25 & 10 \\ 10 & 16 \end{bmatrix}$$

variances

$$\text{var}(x) = 25$$

$$\text{var}(y) = 16$$

covariance

$$\text{cov}(x, y) = 10$$

correlation coefficient

$$r = \frac{10}{\sqrt{25} \sqrt{16}}$$

$$= \frac{10}{20}$$

$$= 0.5$$

$$\rho = 0.5$$

